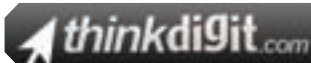


Movies using Lustre

Sin City, Apocalypto, Aeon Flux, Parineeta and Rang De Basanti were some of the movies graded and corrected on the Lustre



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Feature

which are called cans. Each reel is about twenty minutes long, and is first run through the Spirit 4K. The Spirit 4K converts the raw video to one of the highest formats in the industry, which is the 4K format. The Spirit is made by a German company called Digital Film Technology (DFT). If a commercial is brought in, it is processed immediately on the DaVinci 2K controller, which is housed in the same room as the Spirit 4K. The DaVinci 2K is a colour correction tool, and the work for broadcast is done on this machine.

If the client has come in with a feature length film, the reels are run through the Spirit 4K, saved on the servers (more on that later), and colour corrected on one of the two Lustres. The studio has a Lustre 2006 and a Lustre 2010. This is the standard equipment used worldwide, and even Hollywood studios have the same machines. Even highly specialised hardware such as the Lustre machines have beta versions, which were tested out by Peter Jackson while he was working on the Lord of the Rings trilogy. The colour correction artist sits on an elevated stage, with his hands on the Luster, and a Barco projector near his feet projects the video on a screen opposite the artist. All the colour-correction work is done in real time, and in full resolution. This allows a client to sit beside the artist, and not have to worry about render time, which was a common delay to projects till recently.

The video itself is colour-corrected shot by shot, twenty minutes at a time. Either the whole frame is worked on, or a part of the frame. Applying colour-correction to a part of a frame is known as masking, and is useful when there are multiple sources of light, such as a number of windows in an indoor scene. The Lustre allows for as many masks as necessary.

There are three Smoke machines, which are typically used for special effects, transitions and basic animations. The software used is known as Discreet Smoke, and the latest version runs on Linux. The

DR. PRADEEP JAIN ON "THE FINAL GOAL"



"When television came in, it was supposed to be a source of education, but it became an idiot box. When animation came in, it was also supposed to be for education. One of the films we are making, "The Final Goal" is about football. It is more than just an entertainer. One of the interesting things we are doing is that we actually, very subtly, teach the techniques of playing football through animation. When you watch football on television, it is very difficult to see what the players are doing. Through animation, we can show these moves, from whatever angle we want."

Smoke machines are used for packaging or stylizing the video, including adding text and backgrounds. Both the Lustre and the Smoke machines are from Autodesk, which is based in the United States. The new Smoke machines can directly work on the raw Red format. Red cameras are one of the leading cameras in the industry for digital video, and the raw format consumes a lot of space, so this is no mean feat.

There is one Avid machine, which is mostly used for commercials and music videos. The shots are sequenced on the Avid machine, and then saved to be processed by the Lustre or the Smoke machine. Sometimes, the output from the Avid is used directly. Typically, an hour or so of raw video footage (called "rushes") is cut down to about five minutes of edited sequence.

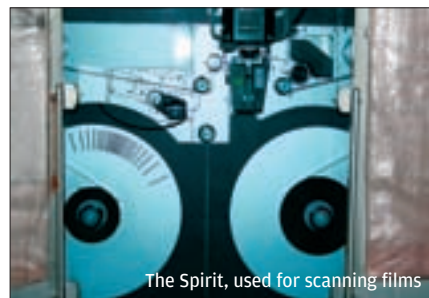
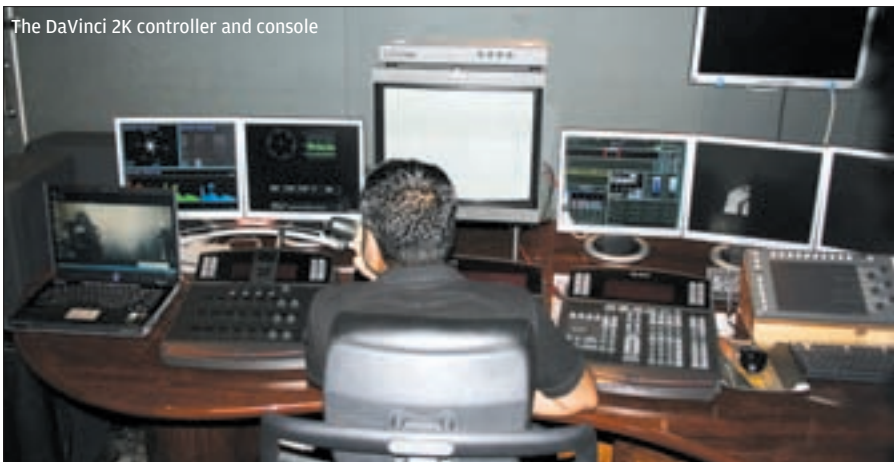
The Ari, another German made machine, is used for the final output. The client may want the output in a number of formats, or in a particular format, which this machine produces. From the film, or open format, the video is downsized to HD format, 35mm, 70mm or broadcast tapes known as digi-betas. It can also print every frame in a video frame by frame, which would give stacks of paper for every reel.

The Ari Machine, and the three smoke machines, as well as the servers are housed in the same room, which is connected by fibre cables to all the other machines in the studio. The storage capacity of the server is

100 terabytes, but all the data is configured on a RAID5 system, which automatically creates a backup of all the data. Essentially, the storage capacity of the server is 50 terabytes, which is sufficient for the studio to work on seven 2K films at the same time. 4K is one of the highest formats in the industry, but is not used regularly as it is not required, and is more time consuming. The servers are designed in a way that there is no need to move or copy and paste data, significantly reducing the time necessary to work for any client. Moving a 300 GB file takes up 4 to 5 hours, which is a considerable amount of time. The servers can handle four streams of real time 2K, which means that both the Lustres can work on the same project, while the Spirit scans data and dumps it on the servers. The throughput of the server is 8 GBps, and 16 GB of RAM is used on all the systems, which is the maximum that is currently possible.

Avitel Post Studioz has worked on many advertisements, films and music videos that you have seen on television. A version of its show reel is on this month's DVD, which shows some of the work that they do. This involves special effects, green screening and animation. The official web site is at www.avitelpoststudioz.com. 

The DaVinci 2K controller and console



The Spirit, used for scanning films



The Lustre 2010, used for grading and colour correction